



Tire Air Permeation Test

1 Scope

Note: Nothing in this standard supercedes applicable laws and regulations.

Note: In the event of conflict between the English and domestic language, the English language shall take precedence.

1.1 Purpose. To determine the rate of air permeation through the liner and carcass of a passenger car or light truck tire.

1.2 Foreword. Tire inflation pressure is monitored during a static soak period up to 180 days in a controlled environment. Data is fitted to a mathematical model and a loss rate of percent per month is determined. It has been shown that this loss rate is relatively unaffected by the instantaneous pressure and tends to normalize results obtained from a variety of tire types and initial test pressures.

1.3 Applicability. Passenger car, compact spare and light truck tires.

2 References

Note: Only the latest approved standards are applicable unless otherwise specified.

2.1 External Standards/Specifications.

None

2.2 GM Standards/Specifications.

GMW16590

2.3 Additional References.

Tire Air Permeability Study

3 Resources

3.1 Facilities. Not applicable.

3.2 Equipment.

3.2.1 Bourdon type pressure gauges with $\pm 1\%$ accuracy over the range of test operation. Gauge scale should be selected so that operation will be in upper half of scale, recommended gauge face should have scale increments of 5 kPa (0.7 psi) or less, and be large enough and clear enough to read or interpolate pressure to the nearest 1 kPa (0.1 psi).

Optional: Inflation pressure transducer with a maximum resolution of 0.3 kPa over the range of test operation.

3.2.2 Reducer coupler for brass fitting, brass valves, approximately 5.08 cm (2 in) long, brass gauge fittings, and copper tubing as required.

3.2.3 Water tank for detecting gross air leakage.

3.2.4 An indoor storage area with the average ambient temperature maintained at $21\text{ }^{\circ}\text{C} \pm 3\text{ }^{\circ}\text{C}$.

3.2.5 Thermometer or other temperature indicating device.

3.2.6 Propane torch and resin core solder.

3.2.7 Barometer.

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3.2.8 Painted steel or aluminum wheels of a specified size meeting Tire and Rim Association (T&RA) dimensional requirements. Rims must have clean, smooth surfaces in the bead seat area, particularly in the vicinity of the weld. Rim flanges must be free of sharp edges or scuffs that could damage the tire during mounting. If other than a painted steel rim is used, the entire rim must be checked for leaks.

3.3 Test Vehicle/Test Piece.

3.3.1 Tires of specified nominal size meeting dimensional requirements of T&RA or European Tyre and Rim Technical Organisation (ETRTO).

3.3.2 A minimum sample size of two of each tire design is recommended.

3.4 Test Time.

Calendar time: 180 days (approximately)

Test time: 180 days plus 48 h setup

Coordination hours: 1 h

Note: 180 days or shorter duration as approved by GM.

Note: **Calendar time** is the overall time from start to finish assuming the parts are available. **Test time** is the actual time for test setup, test execution, and test breakdown. **Coordination hours** is the time for coordinating the test request, part procurement, part modification, analysis, and part disposition.

3.5 Test Required Information. Not applicable.

3.6 Personnel/Skills. Not applicable.

4 Procedure

4.1 Preparation.

4.1.1 Inspect and clean the wheel bead seat area and check both bead seat diameters for compliance with a ball tape.

4.1.2 Drill a 1.58 cm ($\frac{5}{16}$ in) hole in the outside flat surface of the wheel between the rear bead seat and the drop center. File rough edges. Install a 5.08 cm (2 in) brass valve.

4.1.3 A commercial bead rim lubricant shall be applied to the tire bead areas and rim before mounting. Vegetable oil or soap based lubricants are recommended.

4.1.4 Mount tire on wheel using Rubber Manufacturers Association (RMA) recommended practice, but in no case inflate the tire more than 275 kPa (40 psi) to seat the beads.

4.1.5 Calibrate gauge and record data, actual vs. indicated, over a range including 80% to 100% of test pressure with the gauge oriented in the same position as it will be used during the test. Accuracy of $\pm 1\%$ including hysteresis must be maintained in the 80% to 100% test pressure range.

4.1.6 Inflate tire to the maximum or rated pressure and check wheel, fittings, and all bead seat areas for leakage by spraying with leak detection fluid. Correct leaks as required.

4.2 Conditions.

4.2.1 Environmental Conditions. Average ambient temperature: $21\text{ }^{\circ}\text{C} \pm 0.6\text{ }^{\circ}\text{C}$. Daily temperature range: $21\text{ }^{\circ}\text{C} \pm 3\text{ }^{\circ}\text{C}$.

4.2.2 Test Conditions. Deviations from the requirements of this standard shall have been agreed upon. Such requirements shall be specified on component drawings, test certificates, reports, etc.

4.2.2.1 Tires standardized by an organization other than T&RA and ETRTO are to be tested at the same inflation pressure as the equivalent T&RA and ETRTO size.

4.2.2.2 For initial inflation pressures, refer to Appendix A, Table A1.

4.3 Instructions.

4.3.1 Inflate tire to approximately 15 kPa (2 psi) over test pressure to allow for tire growth. Immerse tire assembly, not gauge, into water for approximately 30 minutes to insure that there are no obvious leaks.

4.3.2 After confirming that the tire and rim assembly is free from leaks, the valve should be fitted with a sealing cap, and the tire should be kept in the same orientation to avoid causing new leaks.

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4.3.3 After the leakage check, tires shall be conditioned at the test site ambient temperature for 48 h, and then adjusted to the starting test pressure. If a pressure drop of more than 3 kPa (0.5 psi) has occurred over the conditioning period, the assembly should be rechecked for leakage according to 4.1.6 and, if necessary, remounted. Greater than 48 h conditioning may be necessary for some tires such as high pressure compact spares, whose growth can affect early inflation loss results.

4.3.4 Test tires shall be placed in the controlled temperature environment to allow free air circulation.

If using dial gauges, the tires should be oriented to provide easy visual access to the pressure gauges. Maintain the original orientation of the pressure gauges throughout the test since the orientation of a pressure gauge will influence the reading.

The tires shall not be moved during the test.

4.3.5 Record the starting pressure, date, time, ambient temperature, and barometric pressure.

4.3.6 Read and record all pressure readings to the nearest 1 kPa (0.1 psi) for passenger car tires. Readings with resolution of 2.0 kPa (0.20 psi) are acceptable for light truck tires. Generally, interpolations with resolution finer than one-fifth ($\frac{1}{5}$) of the smallest scale increment are not recommended. If using dial gauges, tap gauge lightly prior to each reading.

4.3.7 Continue monitoring and recording pressure, date, time, barometric pressure, and temperature, as follows at a minimum:

- Once per day for the first two weeks
- Once per week thereafter

Tires shall be considered satisfactorily conditioned when the slope of the logarithm of the normalized pressure versus time relationship becomes constant.

Note: Data variability can be reduced by correcting for temperature and barometric pressure variations during the test. Since the temperature level will also affect permeation rate, an average temperature near 21 °C (70 °F) should be maintained with temperature variation minimized.

4.3.8 Monitor data closely throughout the test, but particularly during the early days and weeks, for evidence of miscellaneous leakage. Comparison to previous data or to other tires in the same test sample is the best reference but, in general, a leak rate of greater than 15 kPa (2 psi) per week should be investigated.

4.3.9 The test shall be continued if the inflation pressure of the other tire in the test sample agrees within 6 kPa (1 psi) inflation pressure after two weeks. Otherwise, the suspect assembly shall be rechecked for leaks according to 4.1.6, and the test restarted.

4.3.10 Continue test to the duration approved by GM or until prediction by the math model suggests an earlier termination. See 5.1 and 5.2.

4.3.11 Disassemble gauge from tire and wheel assembly at the end of test and check and record gauge calibration data as in 4.1.5.

5 Data

5.1 Calculations.

5.1.1 Normalize individual tire pressure measurements for variations of barometric pressure and ambient temperature as follows:

$$P_n = \frac{T_R + 273.2}{T_A + 273.2} (P_{gauge} + BP_{obs}) - BP_{ref}$$

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Where:

p_n	=	Normalized gauge pressure (kPa)
P_{gauge}	=	Observed gauge pressure (kPa)
BP_{obs}	=	Observed barometric pressure (kPa) (or 3.374 × in mercury (at 21 °C))
BP_{ref}	=	Reference barometric pressure (101.3 kPa)
T_R	=	Reference temperature (21 °C)
T_A	=	Ambient temperature in Celsius (°C)

Note: Temperature must be in absolute units for calculations. The preceding equation corrects the temperature to Kelvin. Reference temperature and reference barometric pressure should be typical values for the test environment.

5.1.2 Fit data to a model of the form:

$$p = p_0 e^{bt}$$

Where:

p	=	Instantaneous pressure (kPa)
p_0	=	Initial pressure (kPa)
b	=	Daily loss rate (fraction)
t	=	Time (days)

A least squares fit can be obtained. Eliminate the first 30 days or use the linear portion of the logarithm of the normalized pressure versus time curve, after transformation of the equation to the form:

$$\ln p = A + bt$$

$$A = \ln p_0$$

The model is derived from a relationship that expresses instantaneous pressure loss as a function of instantaneous pressure only,

$$\frac{dp}{dt} = bp$$

Pressure loss in absolute units will vary as the actual nominal pressure changes, but a relative loss rate can be expressed as a constant. The calculated loss rate constant, b , will be in units of 1/day. This number will typically be a very small decimal and it is convenient, and perhaps more intuitively meaningful, to express loss rate as a percent loss per month (multiply by 100% × 30 days/month).

Calculations of loss rate and predictions of future pressures can be made from any point in the test after linearity is achieved (no additional tire growth). The accuracy of such predictions will depend on the quality of the data obtained which, in turn, will depend on factors such as:

- Accuracy in inspection of pressure gauges when taking measurements manually
- Resolution of pressure gauges
- Minimization of temperature variation in environment

5.2 Interpretation of Results. Ninety (90) day pass/fail criteria.

5.2.1 For pass/fail type testing associated with the tire development process, experience with this procedure and equipment has shown that the following rules may be applied for nominal loss rates of 1% to 5% per month.

5.2.1.1 If the best fit calculated loss rate at 90 days is 0.3% below the target or specification, the tire will meet the target or specification at 180 days with a high degree of confidence. The test should continue through 180 days for completion to determine the final performance level for future reference purposes.

5.2.1.2 If the best fit calculated loss rate at 90 days is 0.6% above the target or specification, the tire has small probability of meeting the 180 day permeation requirement and the test may be terminated.

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5.3 Test Documentation. The test results documentation should include:

- Test results report
- Test result data files (extensible markup language (XML) format)
- Data plots

A sample report is given in Appendix B, Report B1.

Note: For data being loaded into the Global Tire-Wheel Test System (GTWTS), submit electronically the test results data files (XML format) and other supporting files listed. GMW16590 defines the electronic file details (e.g., file types, file naming convention, and detailed file formats) for the GTWTS.

5.3.1 Appendix C (Table C1) is an example table of recorded data. For each test tire, record loss rate as a percent per month ($b \times 3000$).

5.3.2 Appendix D (Figure D1), Tire Air Permeability Study, is a sample plot of the logarithm of normalized inflation pressure, normalized pressure and ambient temperature versus time for that tire.

5.3.3 Record any additional test parameters according to local region specifications.

6 Safety

This standard may involve hazardous materials, operations, and equipment. This standard does not propose to address all the safety problems associated with its use. It is the responsibility of the user of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

7 Notes

7.1 Glossary.

Ball Tape: A device available from the Tire and Rim Association for measuring bead seat diameters.

Commercial Truck Tire: A tire developed primarily for use on commercial trucks.

Inflation Pressure Loss Rate: The rate of change of normalized inflation pressure determined from the slope of the linear portion of the log pressure versus time curve.

Light Truck Tire: A tire developed primarily for use on light trucks.

Measured Inflation Pressure: The gauge pressure measured at a given time under ambient temperature and barometric pressure.

Normalized Inflation Pressure: The measured inflation pressure adjusted, according to the ideal gas law, to the nominal test temperature and one atmosphere barometric pressure.

Passenger Car Tire: A tire developed primarily for use on passenger cars.

Test Pressure: The test tire inflation pressure specified for the initiation of the test stabilized at the ambient temperature.

7.2 Acronyms, Abbreviations, and Symbols.

BP	Barometric Pressure
ETRTO	European Tyre and Rim Technical Organisation
GTWTS	Global Tire-Wheel Test System
RMA	Rubber Manufacturers Association
T&RA	Tire and Rim Association
XML	Extensible Markup Language

8 Coding System

This standard shall be referenced in other documents, drawings, etc., as follows:

Test to GMW15000

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9 Release and Revisions

This standard was originated in January 2006. It was first approved by Tire Global Subsystem Leadership Team in July 2006. It was first published in December 2006.

Issue	Publication Date	Description (Organization)
1	DEC 2006	Initial publication.
2	NOV 2011	Five year refresh. Changes to section 5.3 and Appendix B. (Tire GSSLT)
3	JAN 2013	Change to 5.1.1. (Tire and Wheel GSSLT)
4	FEB 2013	Reversion for Template Exception (3.1.1 and 3.1.2 removed). (Tire and Wheel GSSLT)

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Appendix A**Table A1: Test Tire Inflation Pressures**

Tire Type	Load Range	Test Tire Inflation Pressure
Passenger Car	All	250 kPa
Light Truck or Commercial	All	Maximum Inflation Pressure
Compact Spare	All	420 kPa

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Appendix B

Report B1: Sample Test Report (Page 1 of 2)

GMW15000 Air Permeation Report															
Test Request Number: PQMS Activity Number:															
Requester: Requester's Region:															
Model Year / Program: Test Purpose:															
PART DETAILS															
Supplier / Trade Name:				Design Vehicle:				Construction Number:				Tire Size/Load Range:			
Load Index / Load (N):				Tire Type:				Tire Condition:				Sidewall:			
Speed Rating:				Part Status Code:											
Part Numbers:															
TEST CONDITIONS															
TR Comment:												Requested Rim Width (in):			
TEST RESULTS															
Tester / Site:												TIP Approved: Tested Rim Width (in):			
										Normalized Results					
										Loss Rate (%/Mo)				Annualized Loss (kPa)	
Tire ID	DOT	Machine ID	Start Date	Days on Test	Start Press (kPa)	Final Press (kPa)	Temp Range (Deg C)	Temp Avg (Deg C)	90 Day	180 Day	Final	Std Err	90 Day	180 Day	Final
Average for N = :															
Standard Deviation															
GM Proprietary															
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Date Printed:															

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Report B1: Sample Test Report (Page 2 of 2)

GMW15000 Air Permeation Report



Test Request Number:

PQMS Activity Number:

TEST RESULTS

Environmental Conditions

Tire ID	Tester	Start Date	Average Baro Pressure (kPa)	Min Baro Pressure (kPa)	Max Baro Pressure (kPa)	Average Temperature (C)	Min Temperature (C)	Max Temperature (C)	Data corrected to 21 Deg C?	Data corrected to 101 kPa?

Tire ID	Tester	Test Date	Comment	Result Status		
				Approval	Name	Date

FILE ATTACHMENTS

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Date Printed:

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Appendix C

Table C1: Example of Test Data List Record ^{Note}

DATE	AMBIENT TEMP		BAROMETRIC PRESSURE		OBSERVED GAUGE PRESSURE		NORMALIZED GAUGE PRESSURE	DAYS ON TEST	LOSS	PERCENT LOSS
	(F)	(C)	(1Hg)	(kPa)	(PSI)	(kPa)	(kPa)			
04/27/88	67.5	19.7	28.5	96.2	59.2	408.2	405.3	2	0.0	0.0%
05/04/88	71.1	21.7	28.9	97.5	58.9	406.1	401.0	9	4.2	1.0%
05/11/88	71.1	21.7	29.0	97.8	58.5	403.3	398.6	16	6.7	1.7%
05/18/88	70.0	21.1	29.0	97.8	58.1	400.6	396.9	23	8.4	2.1%
05/25/88	71.4	21.9	29.0	97.8	57.8	398.5	393.5	30	11.9	2.9%
06/01/88	70.5	21.4	28.9	97.5	57.8	398.5	394.0	37	11.3	2.8%
06/08/88	70.0	21.1	28.7	96.8	57.4	395.8	391.1	44	14.2	3.5%
06/15/88	70.5	21.4	29.0	97.8	57.0	393.0	399.8	51	16.5	4.1%
06/22/88	70.5	21.4	28.7	96.8	56.9	392.3	387.1	58	18.2	4.5%
06/29/88	70.9	21.6	28.9	97.5	56.6	390.2	385.4	65	19.9	4.9%
07/06/88	70.5	21.4	29.1	98.2	56.4	388.9	385.1	72	20.2	5.0%
07/13/88	70.5	21.4	28.9	97.5	56.1	386.8	382.3	79	23.0	5.7%
07/20/88	71.1	21.7	28.9	97.5	55.9	385.4	380.5	86	24.9	6.1%
07/27/88	70.5	21.4	28.9	97.5	55.7	384.0	379.5	93	25.8	6.4%
08/03/88	70.5	21.4	29.0	97.8	55.4	382.0	377.8	100	27.5	6.8%
08/10/88	72.0	22.2	29.0	97.8	55.3	381.3	375.9	107	29.5	7.3%
08/18/88	71.4	21.9	28.8	97.3	55.0	379.2	373.7	115	31.6	7.8%
08/24/88	70.0	21.1	28.7	96.7	54.8	377.8	373.0	121	32.3	8.0%
08/31/88	70.0	21.2	290.0	97.8	54.8	377.8	374.1	128	31.2	7.7%
09/07/88	70.5	21.4	28.9	97.6	64.3	374.4	370.1	135	35.3	8.7%
09/14/88	71.1	21.7	28.9	97.4	54.0	372.3	367.3	142	38.1	9.4%
09/21/88	70.0	21.1	29.0	97.7	53.7	370.2	36.4	149	38.9	9.6%
09/28/88	70.0	21.1	29.1	98.2	53.4	368.2	364.9	156	40.4	10.0%
10/05/88	71.4	21.9	29.1	98.1	53.4	368.2	363.6	163	41.8	10.3%
10/12/88	69.8	21.0	29.0	97.7	53.2	366.8	363.2	170	42.1	10.4%
10/19/88	70.0	21.2	28.2	97.3	53.1	366.1	361.9	177	43.4	10.7%
10/26/88	70.0	21.1	28.7	96.8	53.0	365.4	360.7	184	44.6	11.0%

Note: ID-144 SUPPLIER CONST NO. DOT = SIZE = T115/70D14 LR-SL TYPE-PSDN

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Appendix D

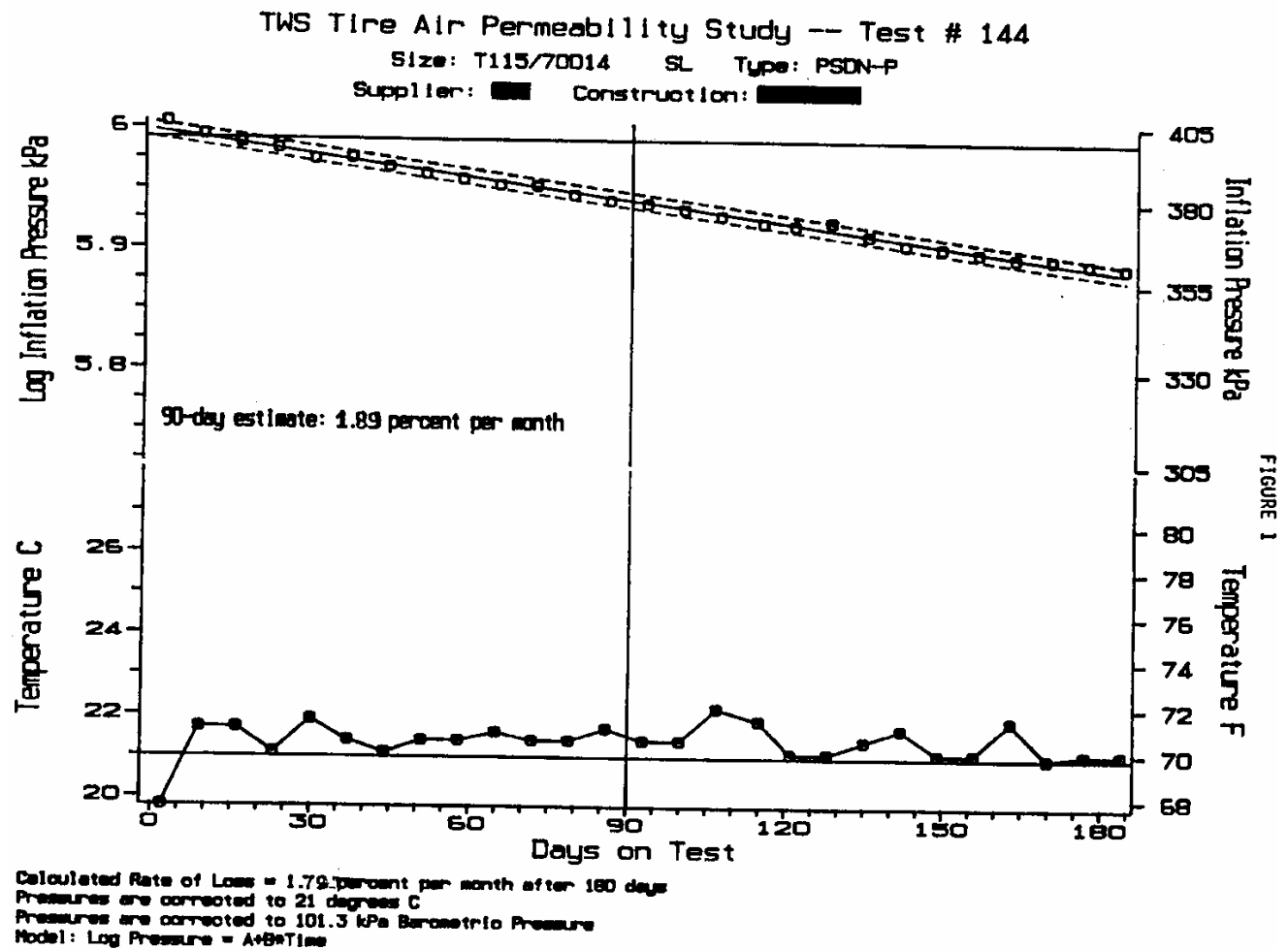


Figure D1: Tire Permeability Study

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